Signed weighted residues of the Fricke Hauptmoduln of levels $2, 3, 5, 7, 13$ modulo primes

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Abstract

For $N \in \{2,3,5,7,13\}$ let $T_N = u_N + a_N + b_N/u_N$ be the normalized Hauptmodul of $\Gamma_0(N)^+$, and for primes $p \geq 5$, $p \neq N$, let $R_{N,p} \in \mathbb{F}_p(T)$ represent $\overline{T_N|U_p}$. We prove that the reduced denominator of $R_{N,p}$ is the product of $T - t_i$ over the *thin* supersingular values (effective inertia $e_i = 1$), that all these poles are simple with nonzero residue, and that

$$\mathrm{Res}_{T=t_i} R_{N,p} \cdot \prod_{k \neq i} (t_i - t_k)^{(p+1)/e_k} = -\left(\frac{-N}{p}\right)$$

at every thin value, rational or not. The proof is uniform in N. A Kaneko–Zagier-type q-expansion argument identifies the residue with the inverse square of the derivative of the Hasse invariant. This reduces the product to $E_{p+1}^{p-1}/(\Delta(\tau)\Delta(N\tau))^{(p^2-1)/24}$ at the supersingular point. Baker's supersingular congruence supplies the factor $-\left(\frac{-1}{p}\right)$. A diamond-operator computation with the multiplier of $\eta(\tau)/\eta(N\tau)$ supplies the factor $\left(\frac{N}{p}\right)$: we show $(\Delta(\tau)/\Delta(N\tau))^{(p^2-1)/24} = \left(\frac{N}{p}\right)$ at every supersingular point of $X_0(N)$, which refines a recent result of Ono. As corollaries, all residues and weighted products lie in $\mathbb{F}_p^\times$, and $E_{p+1}^{(p-1)/2}/\Delta^{(p^2-1)/24}$ is constant on the supersingular locus. We observe numerically, for $p \leq 997$, that this constant equals $\left(\frac{6}{p}\right)\left(\frac{p-1}{2}\right)!$. The proof imports standard facts about mod-p modular forms, moduli of elliptic curves and eta-quotients, and Baker's theorem; these are listed separately. Independent exact computations agree with every step: certified reconstructions of $R_{N,p}$ for all 222 pairs with $p \leq 211$, and a further 960 pairs with $p \leq 1499$ via a proven residue lemma. No exception was found.

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1 Setting and results

1.1 Normalization

Throughout, $p \geq 5$ is prime, $N \in \mathcal{N} := \{2, 3, 5, 7, 13\}$, $p \neq N$, $q = e^{2\pi i\tau}$, and

$$a_N = \frac{24}{N-1}, \qquad b_N = N^{12/(N-1)}, \qquad u = u_N = \Big(\frac{\eta(\tau)}{\eta(N\tau)}\Big)^{a_N}, \qquad T = T_N = u + a_N + \frac{b_N}{u}.$$

The pairs (a_N, b_N) are $(24, 4096), (12, 729), (6, 125), (4, 49), (2, 13)$. From $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$ one gets $u(-1/(N\tau)) = N^{a_N/2}/u(\tau) = b_N/u(\tau)$, so $w_N(u) = b_N/u$ and T is invariant under $\Gamma_0(N)^+ = \langle \Gamma_0(N), w_N \rangle$. Since $u = q^{-1} - a_N + O(q)$, $T = q^{-1} + 0 + O(q)$. The expansions begin

$$T_2 = q^{-1} + 4372q + 96256q^2 + \cdots, \quad T_3 = q^{-1} + 783q + 8672q^2 + \cdots, \quad T_5 = q^{-1} + 134q + 760q^2 + \cdots,$$

$$T_7 = q^{-1} + 51q + 204q^2 + \cdots, \qquad T_{13} = q^{-1} + 12q + 28q^2 + \cdots,$$

the McKay–Thompson series of classes $2A, 3A, 5A, 7A, 13A$ (Conway and Norton, 1979). We checked exactly over $\mathbb{Z}$, to q^{299}, that T has zero constant term, that $j = P_N(u)/u^N$ for the j-maps of the brief,

$$P_2 = (u + 256)^3, \quad P_3 = (u + 27)(u + 243)^3, \quad P_5 = (u^2 + 250u + 3125)^3,$$

$$P_7 = (u^2 + 13u + 49)(u^2 + 245u + 2401)^3,$$

$$P_{13} = (u^2 + 5u + 13)(u^4 + 247u^3 + 3380u^2 + 15379u + 28561)^3,$$

and that $j(N\tau) = P_N(b_N/u)/(b_N/u)^N$ (file `normalization_checks.json`). The last identity confirms $w_N(u) = b_N/u$ algebraically.

For a Laurent series $f = \sum c_n q^n$ put $f|U_p = \sum c_{pn} q^n$. We write $k := (p^2 - 1)/24 \in \mathbb{Z}$, A and B for the reductions of E_{p-1} and E_{p+1} modulo p (both have p-integral q-expansions, and A has q-expansion 1), $\Delta = \eta^{24}$, and $\Delta_N := \Delta(\tau)\Delta(N\tau)$.

1.2 Weights

Let $t_1, \ldots, t_s \in \overline{\mathbb{F}}_p$ be the distinct supersingular values of T. For a point $x = (E, C)$ of $X_0(N)_{\overline{\mathbb{F}}_p}$ over t_i put

$$m_x = \big|\mathrm{Aut}(E, C)/\{\pm 1\}\big|, \qquad e_i = m_x \,\big|\mathrm{Stab}_{\langle w_N \rangle}(x)\big|.$$

Call t_i *thin* if $e_i = 1$ and put $D_{\mathrm{thin}}(T) = \prod_{e_i=1}(T - t_i)$. For thin t_i let

$$r_i = \mathrm{Res}_{T=t_i} R_{N,p}(T), \qquad W_i = \prod_{k \neq i} (t_i - t_k)^{(p+1)/e_k}.$$

Section 4 shows that e_i does not depend on the representative of the Fricke orbit, that it is the effective (tame) inertia of the quotient map at x (not the order of a group containing ± 1), and that every exponent $(p + 1)/e_k$ is an integer.

1.3 Results

Main Theorem. *Let $N \in \{2,3,5,7,13\}$ and let $p \geq 5$ be a prime with $p \neq N$.*

(a) ***(Rationality.)*** *There is a unique $R_{N,p} \in \mathbb{F}_p(T)$ with $\overline{T|U_p} = R_{N,p}(\overline{T})$. It vanishes at $T = \infty$.*

(b) ***(Pole support.)*** *$R_{N,p} = P/D_{\text{thin}}$ with $P \in \mathbb{F}_p[T]$, $\deg P < \deg D_{\text{thin}}$ and $\gcd(P, D_{\text{thin}}) = 1$. Every pole is simple and has a nonzero residue. In particular $\operatorname{den}_{\text{red}} R_{N,p} = D_{\text{thin}}$.*

(c) ***(Residue as a Hasse derivative.)*** *If t_i is thin and x lies over t_i, then*

$$r_i = \frac{144\,(\theta T)(x)^{p+1}}{B(x)^2}, \qquad \theta = q\frac{d}{dq},$$

a ratio of two mod-p modular forms of weight $2(p+1)$, evaluated at x.

(d) ***(Product.)*** *For thin t_i, with $x = (E, C)$ over t_i and $E' = E/C$,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \left(\frac{N}{p}\right) c(E)\, c(E'), \qquad c(E) := \frac{B(E,\omega)^{(p-1)/2}}{\Delta(E,\omega)^k}.$$

(e) ***(Signed weighted residue formula.)*** *For every thin t_i,*

$$\boxed{r_i W_i = -\left(\frac{-N}{p}\right)} \qquad \text{equivalently} \qquad r_i = -\left(\frac{-N}{p}\right) \prod_{k \neq i} (t_i - t_k)^{-(p+1)/e_k}.$$

In particular $r_i \in \mathbb{F}_p^\times$ and $W_i \in \mathbb{F}_p^\times$, also when $t_i \notin \mathbb{F}_p$.

(f) ***(Hecke form.)*** *With $\Psi_{N,p}(X,Y) = (X^p - Y)(X - Y^p) + pH_{N,p}(X,Y)$ as in Proposition 3.1, $\overline{H}_{N,p}(t^p, t) = \left(\frac{-N}{p}\right) W_i^{-1}$ if $t = t_i$ is thin, and $\overline{H}_{N,p}(t^p, t) = 0$ for every other $t \in \mathbb{F}_{p^2}$.*

Theorem 1.1 (Frobenius on the eta quotient)**.** *Let $\ell \neq p$ be prime. At every supersingular point x of $X_0(\ell)_{\overline{\mathbb{F}}_p}$,*

$$\left(\frac{\Delta(\tau)}{\Delta(\ell\tau)}\right)^{(p^2-1)/24}(x) = \left(\frac{\ell}{p}\right).$$

For $\ell = N \in \mathcal{N}$ this reads $u_N(x)^{(p^2-1)/a_N} = \left(\frac{N}{p}\right)$.

Corollary 1.2 (a square root of Baker's congruence)**.** *The value $c(E)$ is the same for all supersingular $E/\overline{\mathbb{F}}_p$. Call it c_p. Then $c_p \in \mathbb{F}_p$ and $c_p^2 = -\left(\frac{-1}{p}\right)$; equivalently*

$$E_{p+1}^{(p-1)/2} \equiv c_p\, \Delta^{(p^2-1)/24} \pmod{(p, E_{p-1})}.$$

Conjecture 1.3 (observed for $5 \leq p \leq 997$)**.** $c_p = \left(\frac{6}{p}\right) \cdot \left(\frac{p-1}{2}\right)! \bmod p$.

Status. Parts (a)–(f), Theorem 1.1 and Corollary 1.2 are proved below, given the imported facts of Section 2. Conjecture 1.3 is supported only by computation. The argument is AI-assisted and has not been refereed. Section 11 lists every step that rests on an import or that a specialist should check.

Where the sign comes from. The constant factors as $-\left(\frac{-N}{p}\right) = \left(-\left(\frac{-1}{p}\right)\right) \cdot \left(\frac{N}{p}\right)$. The first factor is the level-one supersingular sign of de Shalit and Baker; it equals $c(E)^2$. The second factor is $N^{(p-1)/2}$, the degree of the Fricke isogeny raised to the power $(p-1)/2$. It enters through the Kodaira–Spencer factor $B(E') = N\,B(E)$ (Lemma 5.3), equivalently through the eigenvalue $N^{(p-1)/2}$ of w_N on the Hasse invariant. These combine into a constant only because $c(E) = c(E')$. That equality is Theorem 1.1, and its proof uses the Kronecker character of $\mathbb{Q}(\sqrt{N})$ in the eta multiplier. So the suggested mechanism—the determinant N of the Fricke isogeny—is correct, but it explains the factor $\left(\frac{N}{p}\right)$ only together with the eta-quotient Frobenius identity.

Beyond the five levels. The proof uses u_N only through $\Delta/\Delta(N\tau) = u_N^{N-1}$. It applies verbatim to any prime N for which $\Gamma_0(N)^+$ has genus zero, with T the integral normalized Hauptmodul. As an exploratory check we ran the whole certified pipeline for $N = 11$, with $T_{11} = \theta_Q^2/(\eta(\tau)\eta(11\tau))^2 - 6$, $Q = x^2 + xy + 3y^2$, for $5 \le p \le 61$. The constant was $-\left(\frac{-11}{p}\right)$ in every case. These computations are not part of the main claims.

2 Imported facts

The following are taken from the literature and are not re-proved here.

Imported fact I1 (Kronecker congruence)**.** For $\Gamma_0(N)^+$ and its Hauptmodul T with integral q-expansion, the Hecke polynomial $\Psi_{N,p}$ of Proposition 3.1 lies in $\mathbb{Z}[X, Y]$, is monic of degree $p + 1$ in each variable, is symmetric, and satisfies $\Psi_{N,p} \equiv (X^p - Y)(X - Y^p) \pmod p$. The proof of the level-two case in Jordan (2026, Prop. 2.1) goes through verbatim. We sketch it in Proposition 3.1 and check it exactly for $p \le 31$ (Section 8). That manuscript is unpublished. A published reference for general $\Gamma_0(N)^+$ should be cited; we have not verified one.

Imported fact I2 (mod-p modular forms; Katz, 1973; Katz and Mazur, 1985; Serre, 1973)**.** (i) The q-expansion principle holds for mod-p modular forms on $\Gamma_0(N)$, $\Gamma_1(M)$, $p \nmid M$. (ii) $A = E_{p-1} \bmod p$ is the Hasse invariant. It vanishes exactly at the supersingular points and its zeros are simple on the moduli stack (Igusa, 1958). (iii) Supersingular points of $X_0(N)_{\overline{\mathbb{F}}_p}$ are defined over $\mathbb{F}_{p^2}$. (iv) Kodaira–Spencer: on a fine moduli cover, $\omega^{\otimes 2} \cong \Omega^1(\log \text{cusps})$, and a weight-2 form f corresponds to the differential $f\,dq/q$. In particular θT corresponds to dT. (v) Forming coarse quotients commutes with reduction modulo p, because all stabilizers have order prime to p. For $N \in \mathcal{N}$ this is also elementary: u reduces to a coordinate on $X_0(N)_{\mathbb{F}_p} \cong \mathbb{P}^1$, since $j = P_N(u)/u^N$ still has degree $N + 1$ modulo p, and $X_0(N)^+_{\mathbb{F}_p} = X_0(N)_{\mathbb{F}_p}/(u \mapsto b_N/u)$ has coordinate $\overline{T} = u + a_N + b_N/u$, regular off the single cusp.

Imported fact I3 (Baker's supersingular congruence; Baker, 1998, Thm. 1.4)**.** For $p \ge 5$, $E_{p+1}^{p-1} \equiv -\left(\frac{-1}{p}\right) \Delta^{(p^2-1)/12} \pmod{(p, E_{p-1})}$. Equivalently, $B(E, \omega)^{p-1} = -\left(\frac{-1}{p}\right) \Delta(E, \omega)^{(p^2-1)/12}$ for every supersingular E over a finite field. Baker notes that this is Kaneko–Zagier's (40), and hence de Shalit's (39) (Kaneko and Zagier, 1998, §10). We read the statement from the author's online copy and checked it numerically for all $p \le 997$.

Imported fact I4 (eta quotients; Gordon and Hughes, 1993; Newman, 1959; Ono, 2004, Thm. 1.64)**.** Let $f = \prod_{\delta \mid L} \eta(\delta\tau)^{r_\delta}$ with $\sum r_\delta = 0$, $\sum \delta r_\delta \equiv 0$ and $\sum (L/\delta) r_\delta \equiv 0 \pmod{24}$, and $s = \prod \delta^{r_\delta} \in \mathbb{Z}$. Then $f(\gamma\tau) = \left(\frac{s}{d}\right) f(\tau)$ for all $\gamma = \begin{pmatrix} \alpha & \beta \\ c & d \end{pmatrix} \in \Gamma_0(L)$.

Imported fact I5 (moduli and Frobenius; Deligne and Rapoport, 1973; Diamond and Im, 1995; Katz and Mazur, 1985)**.** For $M \ge 5$ and $p \nmid M$, $X_1(M)$ has a smooth model over $\mathbb{Z}[1/M]$. In the μ_M-embedding convention it classifies pairs (E, ι) with $\iota : \mu_M \hookrightarrow E$, and its standard cusp

$(\mathrm{Tate}(q), \mu_M \subset \mathrm{Tate}(q))$ is defined over $\mathbb{Z}[1/M]$. A modular function for $\Gamma_1(M)$ with rational, p-integral q-expansion there is therefore a function on this model defined over $\mathbb{Z}_{(p)}$. A modular unit whose divisor is supported on the cusps, and whose q-expansions at all cusps have unit leading coefficients, is an invertible function on $Y_1(M)_{\mathbb{Z}_{(p)}[\zeta_M]}$. (In the point convention (E, P) the standard cusp is only defined over $\mathbb{Q}(\zeta_M)$, so a function with rational q-expansion need not be defined over $\mathbb{Q}$.) The diamond operator $\langle d\rangle : (E, \iota) \mapsto (E, \iota \circ [d])$ acts on such functions as $f \mapsto f \circ \gamma$ for $\gamma \in \Gamma_0(M)$ with lower-right entry $\equiv d^{\pm 1} \pmod M$; the sign of the exponent depends on conventions and does not matter below. The map $f(\tau) \mapsto f(n\tau)$ corresponds to the usual degeneracy map.

Imported fact I6 (Deuring models; e.g. Ono, 2025, Lemma 4.3)**.** Every supersingular $j \in \overline{\mathbb{F}}_p$ has a model $E/\mathbb{F}_{p^2}$ whose p^2-Frobenius endomorphism is $[-p]$. For such E, every finite subgroup of order prime to p is $\mathbb{F}_{p^2}$-rational, and every curve isogenous to E by an isogeny of degree prime to p again has a model with Frobenius $[-p]$.

Imported fact I7 (connectivity; Mestre, 1986; Pizer, 1990)**.** For a prime $\ell \neq p$, the ℓ-isogeny graph on supersingular j-invariants over $\overline{\mathbb{F}}_p$ is connected.

3 Rationality and the Kronecker correction

Proposition 3.1. *Let Δ_p be the set of integral matrices of determinant p with lower-left entry divisible by N. There is a unique $\Psi_{N,p} \in \mathbb{Z}[X, Y]$ with*

$$\Psi_{N,p}(X, T(\tau)) = (X - T(p\tau)) \prod_{b=0}^{p-1} \Big(X - T\big(\tfrac{\tau+b}{p}\big)\Big).$$

It is monic of degree $p+1$ in each variable, symmetric, and satisfies $\Psi_{N,p} \equiv (X^p - Y)(X - Y^p) \pmod p$.

Sketch, as in Jordan, 2026, Prop. 2.1. The matrices $\mathrm{diag}(p, 1)$ and $\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$, $0 \le b < p$, represent $\Gamma_0(N)\backslash\Delta_p$. Conjugation by $W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$ maps $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ to $\begin{pmatrix} d & -c/N \\ -Nb & a \end{pmatrix}$, so it preserves Δ_p. Hence the elementary symmetric functions of the roots are $\Gamma_0(N)^+$-invariant, holomorphic on the upper half-plane and integral at the cusp. Subtracting integral powers of T puts them in $\mathbb{Z}[T]$. Modulo $1 - \zeta_p$ all $T((\tau + b)/p)$ coincide with $T(q^{1/p})$, whose p-th power is $T(q)$, and $T(p\tau) \equiv T^p$; this gives the congruence. For symmetry: $\Gamma_0(N)^+$ acts transitively on its cosets in $\Gamma_0(N)^+\Delta_p\Gamma_0(N)^+$, so $\Psi_{N,p}(X, T)$ is irreducible and is the minimal polynomial of $T(p\tau)$ over $\mathbb{C}(T)$. Also $\mathrm{diag}(1, p) \in \Delta_p$, so $\Psi_{N,p}(T(\tau), T(p\tau)) = 0$. Hence $\Psi_{N,p}(Y, X) = c\,\Psi_{N,p}(X, Y)$ with $c = \pm 1$. Reducing modulo p, where both sides equal the symmetric $(X^p - Y)(X - Y^p)$, forces $c = 1$. □

Define $H = H_{N,p} \in \mathbb{Z}[X, Y]$ by $\Psi_{N,p} = (X^p - Y)(X - Y^p) + pH$; H is symmetric. Put

$$\psi = \frac{T(q)^p - T(q^p)}{p} \in \mathbb{Z}((q)),$$

which is integral because $T(q^p) \equiv T(q)^p \pmod p$.

Proposition 3.2 (residue criterion)**.** *In $\mathbb{F}_p((q))$,*

$$\overline{\psi} = \frac{\overline{H}(T^p, T)}{T^{p^2} - T}, \qquad \overline{T|U_p} = \overline{Q}(T) + \overline{\psi},$$

where $\overline{Q} \in \mathbb{F}_p[T]$ *is defined by* $-[X^p]\Psi_{N,p} = Y^p + pQ(Y)$. *Hence* $R_{N,p} = \overline{Q}(T) + \overline{H}(T^p, T)/(T^{p^2} - T)$ *exists and is unique. It vanishes at* ∞, *its poles are simple and lie in* $\mathbb{F}_{p^2}$, *and for* $t \in \mathbb{F}_{p^2}$

$$\operatorname{Res}_{T=t} R_{N,p} = -\overline{H}(t^p, t).$$

Proof. This is Jordan (2026, Thm. 2.2) with $E = -\psi$. Substitute $X = T(q^p) = T^p - p\psi$ and $Y = T$ into $\Psi_{N,p} = 0$. Since $X - Y^p = -p\psi$, division by p and reduction give $-\overline{\psi}\,(T^{p^2} - T) + \overline{H}(T^p, T) = 0$. The sum of the roots of $\Psi_{N,p}(X, T)$ is $T(q^p) + p\,(T|U_p) = T^p + pQ(T)$, so $T|U_p = Q(T) + \psi$. The polynomial $T^{p^2} - T$ is squarefree, its roots are exactly $\mathbb{F}_{p^2}$, and its derivative is -1. $R_{N,p}$ vanishes at ∞ because $T|U_p \in q\mathbb{Z}[[q]]$ (the constant term of T is 0). Uniqueness holds because $\overline{T}$ is transcendental over $\mathbb{F}_p$. □

Corollary 3.3. *For every* $t \in \mathbb{F}_{p^2}$, $\operatorname{Res}_t R_{N,p} \in \mathbb{F}_p$ *and* $\operatorname{Res}_{t^p} R_{N,p} = \operatorname{Res}_t R_{N,p}$.

Proof. $(\operatorname{Res}_t R)^p = \operatorname{Res}_{t^p} R = -\overline{H}(t^{p^2}, t^p) = -\overline{H}(t, t^p) = -\overline{H}(t^p, t) = \operatorname{Res}_t R$, using $t^{p^2} = t$, the $\mathbb{F}_p$-rationality of R and the symmetry of H. □

Proposition 3.2 gives part (a) of the Main Theorem and the a priori bound $\operatorname{den}_{\mathrm{red}} R_{N,p} \mid T^{p^2} - T$ used by the certificates in Section 8.

4 Weights and the weighted supersingular polynomial

Lemma 4.1 (effective inertia)**.** *Let* $x = (E, C)$ *be a non-cuspidal point of* $X_0(N)_{\overline{\mathbb{F}}_p}$, *supersingular or ordinary. Let* G_x *be the group of automorphisms of* (E, C) *together with the isomorphisms* $(E, C) \to w_N(E, C) = (E/C, E[N]/C)$ *when* $w_N(x) = x$. *It is a group once* $w_N^2(E, C) \cong (E, C)$ *is identified via* $[N]$.

(i) $|G_x/\{\pm 1\}| = m_x |\operatorname{Stab}_{\langle w_N \rangle}(x)| = e_x$. *The value is the same at* x *and at* $w_N(x)$.

(ii) $\{\pm 1\}$ *is exactly the kernel of the action of* G_x *on the deformation space of* x. *Since* $e_x \in \{1, 2, 3, 4, 6\}$ *is prime to* p, $G_x/\{\pm 1\}$ *is cyclic and acts faithfully on the tangent line. Hence, for a local parameter* z *of a fine moduli cover at a point over* x, $T - t = z^{e_x} \cdot (\text{unit})$.

(iii) *If* x *is supersingular,* $(p + 1)/e_x \in \mathbb{Z}$.

Proof. (i) If $w_N(x) = x$, the isomorphisms $(E, C) \to w_N(E, C)$ form a torsor under $\operatorname{Aut}(E, C)$, so $|G_x| = 2|\operatorname{Aut}(E, C)|$. Conjugation by the isogeny $E \to E/C$ identifies $\operatorname{Aut}(E, C)$ with $\operatorname{Aut}(E/C, E[N]/C)$, so $m_x = m_{w_N(x)}$. (ii) $[-1]$ acts trivially on deformations. Suppose $g \in G_x$, $g \neq \pm 1$, acted trivially. Then g would extend to the universal deformation. Its generic fibre has endomorphism ring $\mathbb{Z}$, and g corresponds either to a nontrivial automorphism or to an endomorphism of degree N, which is not a square. That is impossible. The same argument works at ordinary points. A finite group of order prime to p acting faithfully on $k[[z]]$ is cyclic and acts faithfully on $(z)/(z^2)$, so the invariant ring is $k[[z^e]]$. (iii) $e = 3$ or 6 forces $m_x = 3$, so $j = 0$ is supersingular and $p \equiv 2 \pmod 3$; $e = 4$ forces $m_x = 2$, so $j = 1728$ is supersingular and $p \equiv 3 \pmod 4$. For $p \geq 5$, $m_x \in \{1, 2, 3\}$. □

In the computations m_x is read off from the multiplicity μ of $u(x)$ as a root of $P_N(u) - j\,u^N$: $m_x = 3/\mu$ over $j = 0$, $m_x = 2/\mu$ over $j = 1728$, and $m_x = 1$ (with $\mu = 1$) otherwise. These are ramification indices of the coarse map $X_0(N) \to X(1)$. The stabilizer is detected by $u(x)^2 = b_N$. Two independent checks confirm the weights: the identity of Proposition 4.3, and the vanishing orders of θT predicted in Lemma 5.2.

Lemma 4.2. *As mod-*p *forms on* $\Gamma_0(N)$: $A|_{p-1}W_N = \left(\frac{N}{p}\right) A$, $\Delta_N|_{24}W_N = \Delta_N$ *and* $\theta T|_2 W_N = \theta T$.

Proof. For a level-one form f of weight k, $f|_k W_N = N^{k/2} f(N\tau)$. The form $E_{p-1}(N\tau)$ has q-expansion $\equiv 1 \equiv E_{p-1}(\tau)$, so it equals A modulo p by Fact I2(i); and $N^{(p-1)/2} \equiv \left(\frac{N}{p}\right)$. The other two identities hold in characteristic 0: $\Delta(-1/(N\tau))\Delta(-1/\tau) = N^{12}\tau^{24}\Delta(N\tau)\Delta(\tau)$, and T is w_N-invariant. □

Proposition 4.3 (weighted supersingular polynomial)**.** *Let $S(T) = \prod_{k=1}^{s}(T - t_k)^{(p+1)/e_k}$. Then $S \in \mathbb{F}_p[T]$ and*

$$S(\overline{T}) = \Delta_N^{-k} = q^{-(N+1)k} \prod_{n\geq 1} (1-q^n)^{-24k}(1-q^{Nn})^{-24k} \quad \text{in } \mathbb{F}_p((q)).$$

Equivalently, $S = A^{p+1}/\Delta_N^k$ as functions on $X_0(N)^+_{\mathbb{F}_p}$. In particular $\sum_k e_k^{-1} = (N+1)(p-1)/24$.

Proof. $F := A^{p+1}/\Delta_N^k$ is a ratio of mod-p forms of weight p^2-1. By Lemma 4.2 it is w_N-invariant, since $\left(\frac{N}{p}\right)^{p+1} = 1$. Δ_N does not vanish on $Y_0(N)$, so F is regular off the cusp of $X_0(N)^+$. Its q-expansion there is $\Delta_N^{-k} = q^{-(N+1)k}(1+O(q))$, because A has q-expansion 1. Hence F is a monic polynomial in T of degree $(N+1)k$. By Fact I2(ii) and Lemma 4.1(ii), F vanishes exactly at the supersingular t_k, to order $(p+1)/e_k$ in $T - t_k$. Comparing degrees gives the mass identity. □

5 The Kaneko–Zagier step

The following is the level-N version of Kaneko and Zagier (1998, §10, (35)–(38)); Kaneko and Zagier credit the idea to Koike (1973).

Lemma 5.1. *In $\mathbb{F}_p((q))$, with $d/dT = (\theta T)^{-1}\theta$,*

$$\frac{d\overline{\psi}}{dT} = T^{p-1} - (\theta T)^{p-1}, \qquad (\theta T)^{p-1} = g(\overline{T}), \quad g := \frac{(\theta T)^{p-1}}{A^2} \in \mathbb{F}_p(T).$$

Proof. Over $\mathbb{Z}$, $\theta\psi = T^{p-1}\theta T - (\theta T)(q^p)$, and $(\theta T)(q^p) \equiv (\theta T)^p$. Divide by $\theta T = -q^{-1} + \cdots$, which is invertible. The function g is a ratio of mod-p forms of weight $2(p-1)$, it is w_N-invariant by Lemma 4.2, and it has q-expansion $(\theta T)^{p-1}$ because $A \equiv 1$. □

Lemma 5.2 (local structure of g)**.** *Let $t \in \mathbb{F}_{p^2}$ be a non-cuspidal point with effective inertia e_t, and put $\varepsilon_t = 1$ if t is supersingular and 0 otherwise. Then*

$$\operatorname{ord}_{T=t} g = \frac{(p-1)(e_t-1) - 2\varepsilon_t}{e_t}.$$

For thin t and x over t, let $a = A/(\theta T)^{(p-1)/2}$, a function on $X_0(N)$ near x. Then a has a simple zero at x,

$$a'(t) := \frac{da}{dT}(x) = \frac{(B/12)(x)}{(\theta T)(x)^{(p+1)/2}} \neq 0, \qquad g = a'(t)^{-2}(T-t)^{-2}(1+O(T-t)).$$

Proof. By Fact I2(iv) and Lemma 4.1(ii), θT vanishes to order $e_t - 1$ in a stack parameter z, and A to order ε_t; also $T - t \sim z^{e_t}$. This gives the order formula. For thin t, the map $X_0(N) \to X_0(N)^+$ is étale at x and $\theta T(x) \neq 0$. In characteristic 0, for forms F and M of equal weight, $\theta(F/M) = (\partial F \cdot M - F \cdot \partial M)/M^2$, where $\partial = \theta - \frac{k}{12}E_2$ is the Serre derivative. Apply this with $F = E_{p-1}$ and $M = (\theta T)^{(p-1)/2}$ and reduce modulo p; at x we have $A(x) = 0$. The form ∂E_{p-1} has weight $p+1$ and q-expansion $\equiv \theta(1) + \frac{1}{12}E_2 \equiv \frac{1}{12}E_{p+1}$, using the Kummer congruence $2(p+1)/B_{p+1} \equiv 24$ and $\sigma_p \equiv \sigma_1$. So $\partial A = B/12$ as mod-p forms. Finally $da/dT = \theta a/\theta T$. The value $B(x) \neq 0$ because $B/12$ is the derivative of A at its simple zero. □

Proof of parts (b) and (c) of the Main Theorem. Let $t \in \mathbb{F}_{p^2}$. By Proposition 3.2, near t we have $\overline{\psi} = -\overline{H}(t^p,t)(T-t)^{-1} + O(1)$, because $T^{p^2} - T = -(T-t)(1+O(T-t))$. So the coefficient of $(T-t)^{-2}$ in $d\overline{\psi}/dT$ is $\overline{H}(t^p,t)$. By Lemma 5.1 it is also $-[g]_{-2}$. Hence

$$\operatorname{Res}_t R_{N,p} = -\overline{H}(t^p,t) = [g]_{-2}.$$

By Lemma 5.2, $\operatorname{ord}_t g \geq 0$ unless t is thin. For $p \geq 5$ and $e_t \geq 2$ we have $(p-1)(e_t-1) - 2 \geq 0$. At thin t, $[g]_{-2} = a'(t)^{-2} = 144(\theta T)(x)^{p+1}/B(x)^2 \neq 0$. Every pole is simple by Proposition 3.2, so $R_{N,p} = P/D_{\text{thin}}$ with $P(t_i) \neq 0$ at every thin t_i. Since $R_{N,p}(\infty) = 0$, $\deg P < \deg D_{\text{thin}}$. □

Lemma 5.3 (Kodaira–Spencer factor of the Fricke isogeny)**.** *Let N be any prime $\neq p$ (not only $N \in \mathcal{N}$). Let $x = (E,C)$ be a supersingular point of $X_0(N)$, $\pi : E \to E' = E/C$, and $\omega = \pi^*\omega'$. Then $B(E',\omega') = N\,B(E,\omega)$, $A(E',\omega') = A(E,\omega)$ and $\Delta(N\tau)(x) = N^{-12}\Delta(E',\omega')$.*

Proof. For a level-one form f of weight w, $f(N\tau)$ at (E,C,ω) equals $N^{-w}f(E',\omega')$. Check this on $(\mathrm{Tate}(q),\mu_N)$, where $E' \cong \mathrm{Tate}(q^N)$ and $\pi^*\omega'_{\text{can}} = N\omega_{\text{can}}$. The form $E_2(\tau) - NE_2(N\tau)$ is a holomorphic weight-2 form on $\Gamma_0(N)$ with p-integral coefficients. Modulo p, the forms $B(\tau) - NB(N\tau)$ and $A\cdot(E_2 - NE_2(N\tau))$ have weight $p+1$ and the same q-expansion, so they are equal. At a supersingular point this gives $B(E,\omega) = N\cdot N^{-(p+1)}B(E',\omega') = N^{-1}B(E',\omega')$, because $N^{p-1} = 1$. The statement about A follows in the same way from $A(N\tau) = A(\tau)$. □

Proposition 5.4 (part (d) of the Main Theorem)**.** *For thin t_i and $x = (E,C)$ over t_i,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \beta(E)\Big(\frac{\Delta(\tau)}{\Delta(N\tau)}\Big)^k(x) = \left(\frac{N}{p}\right)c(E)c(E'), \qquad \beta(E) := \frac{B(E,\omega)^{p-1}}{\Delta(E,\omega)^{2k}} = c(E)^2.$$

Proof. Near x, Proposition 4.3 gives $S(T) = A^{p+1}/\Delta_N^k = a^{p+1}\,Q^k$ with $Q := (\theta T)^{12}/\Delta_N$, a function that is regular and nonzero at x. Hence $W_i = [S/(T-t_i)^{p+1}]_{T=t_i} = a'(t_i)^{p+1}Q(x)^k$. With $r_i = a'(t_i)^{-2}$ this gives

$$r_i W_i = a'(t_i)^{p-1}Q(x)^k = \Big(\frac{B}{12}\Big)^{p-1}(\theta T)^{-\frac{p^2-1}{2}}\cdot\frac{(\theta T)^{\frac{p^2-1}{2}}}{\Delta_N^k} = \frac{B^{p-1}}{\Delta_N^k}(x).$$

Write $\Delta_N^k = \Delta^{2k}(\Delta(N\tau)/\Delta)^k$ for the second equality. For the third, Lemma 5.3 gives $\Delta_N(x) = N^{-12}\Delta(E,\omega)\Delta(E',\omega')$ with $N^{12k} = N^{(p^2-1)/2} = 1$, and $B(E)^{(p-1)/2} = N^{-(p-1)/2}B(E')^{(p-1)/2} = \left(\frac{N}{p}\right)B(E')^{(p-1)/2}$. □

6 The eta quotient at supersingular points

Lemma 6.1 (multiplier)**.** *Let ℓ be prime and $y = \eta(\tau)/\eta(\ell\tau)$. For $\gamma = \begin{pmatrix}\alpha & \beta\\ c & d\end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ with $24\ell \mid c$ and $24 \mid \beta$,*

$$y(\gamma\tau) = \left(\frac{\ell}{d}\right)y(\tau).$$

Proof. Apply Fact I4 to $G(\tau) := 1/y(24\tau) = \eta(24\ell\tau)/\eta(24\tau)$ with $L = 576\ell$. Here $r_{24\ell} = 1$, $r_{24} = -1$, $\sum \delta r_\delta = 24(\ell-1)$, $\sum (L/\delta)r_\delta = 24(1-\ell)$ and $s = \ell$. With $\tilde{\gamma} = \begin{pmatrix}\alpha & \beta/24\\ 24c & d\end{pmatrix} \in \Gamma_0(576\ell)$ we have $G(\tilde{\gamma}\sigma) = \left(\frac{\ell}{d}\right)G(\sigma)$, and $\gamma\tau/24 = \tilde{\gamma}(\tau/24)$. □

We checked Lemma 6.1 independently from Rademacher's Dedekind-sum formula for the eta multiplier. For every $\ell \in \mathcal{N}$ and every unit class δ mod 24ℓ, with three lifts $\gamma \equiv \mathrm{diag}(\delta,\delta^{-1})$ each, the multiplier is exactly $\left(\frac{\ell}{\delta}\right)$. Separately, the exact formula agrees with 60-digit numerics on 30 matrices (`eta_multiplier_checks.json`). The character $m \mapsto \left(\frac{\ell}{m}\right)$ is the Kronecker character of $\mathbb{Q}(\sqrt{\ell})$, which is even.

Proof of Theorem 1.1. Put $M = 576\ell$ and $G = 1/y(24\tau)$, so $G^{24}(\tau) = \Delta(24\ell\tau)/\Delta(24\tau)$. By Lemma 6.1, G is a modular unit for $\Gamma_1(M)$ with integral q-expansion $q^{\ell-1}\prod(1-q^{24\ell n})(1-q^{24n})^{-1}$, and $G|\langle d\rangle = \left(\frac{\ell}{d}\right) G$. By Fact I5 (in the μ_M-embedding convention), G is an invertible function on the non-cuspidal part of $X_1(M)$ over $\mathbb{Z}_{(p)}$, hence on $Y_1(M)_{\mathbb{F}_p}$, defined over $\mathbb{F}_p$. Let $x = (E, C)$ be supersingular. By Fact I6 we may take $E/\mathbb{F}_{p^2}$ with Frobenius $[-p]$. The degeneracy map $X_1(M) \to X_0(\ell)$ belonging to $\tau \mapsto 24\tau$ is a finite surjective map of proper curves, so we can choose $\tilde{x} = (E_1, \iota)$ over x. Its curve E_1 is 24-isogenous to E, so by Fact I6 we may take it to be a Deuring model. The p^2-power Frobenius σ acts on μ_M by $\zeta \mapsto \zeta^{p^2}$ and on $E_1(\overline{\mathbb{F}}_p)$ by $\pi_{E_1} = [-p]$, so $\sigma(\tilde{x}) = (E_1, [-p] \circ \iota \circ [p^{-2}]) = \langle -p^{-1}\rangle \tilde{x}$. (In the point convention G is only defined over $\mathbb{Q}(\sqrt{\ell})$, and its reduction over $\mathbb{F}_{p^2}$. Since only the p^2-power Frobenius enters, the same computation with $\langle -p\rangle$ gives the same result.) Since G is defined over $\mathbb{F}_p$,

$$G(\tilde{x})^{p^2} = G(\sigma\tilde{x}) = \left(\frac{\ell}{-p^{\mp 1}}\right) G(\tilde{x}) = \left(\frac{\ell}{p}\right) G(\tilde{x}),$$

because the character $m \mapsto \left(\frac{\ell}{m}\right)$ is even and quadratic. Now $G(\tilde{x})^{24} = (\Delta(\tau)/\Delta(\ell\tau))(x)^{-1}$, so $(\Delta/\Delta(\ell\tau))(x)^{(p^2-1)/24} = G(\tilde{x})^{-(p^2-1)} = \left(\frac{\ell}{p}\right)$. For $\ell = N \in \mathcal{N}$, $\Delta/\Delta(N\tau) = u_N^{N-1}$ and $(N-1)/24 = 1/a_N$. □

Remark 6.2. Ono (2025, Thm. 1.2) proves $R_\ell^{(p^2-1)/12} = 1$ at supersingular points, where $R_\ell = \Delta(\ell\tau)/\Delta(\tau)$ is the inverse of our ratio; inverting does not change a value in $\{\pm 1\}$. His Remark 4 shows that the $(p^2-1)/24$ power can equal -1. Theorem 1.1 determines that power: it is $\left(\frac{\ell}{p}\right)$. Ono's own example ($p = 11$, $\ell = 2$, values $6, 7, 2$, each with fifth power -1) agrees, since $\left(\frac{2}{11}\right) = -1$. Ono's method descends η^8 and η^6 to $\mathbb{F}_{p^2}$. The diamond-operator argument above needs no such descent, because it computes the Frobenius action through the multiplier.

7 Proof of the Main Theorem; corollaries

Proof of part (e). By Proposition 5.4, $r_i W_i = \beta(E)\,(\Delta/\Delta(N\tau))^k(x)$. Fact I3 gives $\beta(E) = -\left(\frac{-1}{p}\right)$, and Theorem 1.1 gives $(\Delta/\Delta(N\tau))^k(x) = \left(\frac{N}{p}\right)$. So $r_i W_i = -\left(\frac{-1}{p}\right)\left(\frac{N}{p}\right) = -\left(\frac{-N}{p}\right)$. By Corollary 3.3, $r_i \in \mathbb{F}_p^\times$, and therefore $W_i = -\left(\frac{-N}{p}\right) r_i^{-1} \in \mathbb{F}_p^\times$. □

Proof of part (f). By Proposition 3.2 and part (b), $\overline{H}(t^p, t) = -\operatorname{Res}_t R_{N,p}$. This is 0 unless t is thin, and equals $-r_i = \left(\frac{-N}{p}\right) W_i^{-1}$ if $t = t_i$ is thin. This fixes the sign: with $\Psi = (X^p - Y)(X - Y^p) + pH$, the residue is $-\overline{H}(t^p, t)$, not $+\overline{H}$. □

Proof of Corollary 1.2. Combining Lemma 5.3 with Theorem 1.1 gives $c(E')/c(E) = \left(\frac{\ell}{p}\right)(\Delta/\Delta(\ell\tau))^k(x) = 1$ for every supersingular ℓ-isogeny $E \to E'$. By Fact I7, c is constant. Fact I3 gives $c^2 = \beta = -\left(\frac{-1}{p}\right)$. Since $-\left(\frac{-1}{p}\right)$ is -1 for $p \equiv 1 \pmod 4$ and $+1$ for $p \equiv 3 \pmod 4$, $c \in \mathbb{F}_p$ in both cases. For the congruence: the level-one mod-p form $B^{(p-1)/2} - c\,\Delta^{(p^2-1)/24}$ vanishes on the supersingular locus. The zeros of A there are simple on the stack, including at $j = 0, 1728$. Hence A divides it in $M_*(\mathbb{F}_p) = \mathbb{F}_p[E_4, E_6]$ ($p \geq 5$).

Remark 7.1. Given Lemma 5.3, Theorem 1.1 for all primes ℓ is equivalent to the constancy of c. It therefore follows from Theorem 1.1 for a single ℓ together with Fact I7.

□

Remark 7.2 (on Conjecture 1.3). By Wilson's theorem $(\frac{p-1}{2})!^2 \equiv (-1)^{(p+1)/2} = -\left(\frac{-1}{p}\right)$, so the conjecture is consistent with Corollary 1.2. For $p \equiv 3 \pmod 4$, Mordell's congruence $(\frac{p-1}{2})! \equiv (-1)^{(h(-p)+1)/2}$ (Mordell, 1961) turns it into $c_p = \left(\frac{6}{p}\right)(-1)^{(h(-p)+1)/2}$. A proof would

follow from evaluating $c(E)$ at a single convenient supersingular curve (for example $j = 1728$ when $p \equiv 3 \pmod 4$); we have not done this.

Remark 7.3 ($\mathbb{F}_p$-rationality at non-rational points). The brief records the observation that r_i and W_i lie in $\mathbb{F}_p$ even when $t_i \notin \mathbb{F}_p$. The two halves have different explanations. $r_i \in \mathbb{F}_p$ is elementary (Corollary 3.3: symmetry of H and Frobenius-equivariance of residues). $W_i \in \mathbb{F}_p$ is not formal: writing $W_i = \mathrm{Nm}(D'_{\mathrm{thin}}(t_i)) \cdot \prod_{e_k>1}(t_i - t_k)^{(p+1)/e_k}$, it amounts to the product $\prod_{e_k>1}(t_i - t_k)^{(p+1)/e_k}$ lying in $\mathbb{F}_p$. Equivalently, a product of power-residue characters (one of order e_k for each factor) must be trivial; this is not visible factor by factor. Here it follows only from part (e).

8 Computations

All computations are exact (python-flint 0.9.0 for $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$ arithmetic; integer and rational arithmetic otherwise). The code and machine-readable output accompany this note; the README explains how to reproduce them. No computation is a premise of any proof.

8.1 Pipeline

(1) *Supersingular j-values* are computed twice: as $256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ over the roots of Deuring's polynomial in $\mathbb{F}_{p^2}$, and as the roots of the Hasse invariant of $y^2 = x^3 + 3j(1728 - j)x + 2j(1728 - j)^2$ together with tests at $j = 0, 1728$. The two sets agreed for every p.

(2) *Points and weights.* For each supersingular j the roots of $P_N(u) - ju^N$ are found in $\mathbb{F}_{p^2}$. They always split completely, which confirms Fact I2(iii). Multiplicities give m_x, $u^2 = b_N$ detects Fricke-fixed points, and T-values are grouped into Fricke orbits; the code asserts orbit structure $\{u, b_N/u\}$ and equal m_x along orbits. Mass identity and integrality of $(p + 1)/e_k$ are checked.

(3) *Certified reconstruction of $R_{N,p}$ (Tier 1).* From the q-expansion of T modulo p to q^{p^3} we form $\overline{T|U_p}$ to q^{p^2}, multiply by $D_{\mathrm{thin}}(\overline{T})$, and read off P of degree $< d = \deg D_{\mathrm{thin}}$ from the principal part. The certificate is that $D_{\mathrm{thin}}(T)\,\overline{T|U_p} - P(T)$ vanishes through q^{p^2-d}. Since $\mathrm{den}_{\mathrm{red}}\, R_{N,p} \mid T^{p^2} - T$ (Proposition 3.2) and $D_{\mathrm{thin}} \mid T^{p^2} - T$, the difference $R_{N,p} - P/D_{\mathrm{thin}}$ has the form $A/(T^{p^2} - T)$ with $\deg A < p^2$. Its expansion would then have a nonzero term of order at most $p^2 - d$ unless $A = 0$. So the certificate proves $R_{N,p} = P/D_{\mathrm{thin}}$ exactly. The candidate denominator uses only the supersingular data, never the conjectured residues. For $p \le 37$ we also compute $\mathrm{den}_{\mathrm{red}}\, R_{N,p}$ with no candidate at all, as $(T^{p^2} - T)/\gcd((T^{p^2} - T)R, T^{p^2} - T)$.

(4) *Residues and products.* $r_i = P(t_i)/D'_{\mathrm{thin}}(t_i)$ and W_i are computed in $\mathbb{F}_{p^2}$. Rationality in $\mathbb{F}_p$, constancy of $r_i W_i$, its value, and the Frobenius relations $r_{t^p} = r_t^p$, $W_{t^p} = W_t^p$ are tested.

(5) *Tier 2 (larger p).* Residues from Lemma 5.2 and the proof of part (b): $r_t = [g]_{-2} = \mathrm{Poly}(t)/D'_{\mathrm{thin}}(t)^2$, where $\mathrm{Poly} = (\theta T)^{p-1} D_{\mathrm{thin}}(T)^2$ is recovered from q-series and verified to 100 further coefficients. This tier depends on the proven Kaneko–Zagier lemma, not on the conjecture. Tier 2 agrees with Tier 1 wherever both were run.

(6) *Proof ingredients.* Proposition 4.3 (exact equality in $\mathbb{F}_p[T]$, $p \le 211$); the double-pole identity $r_t = [g]_{-2}$ and the predicted orders of g at non-thin points; Theorem 1.1 at *every* supersingular point; Baker's congruence; constancy of $c(E)$; and $r_i W_i = \left(\frac{N}{p}\right) c(E)c(E')$.

(7) *Hecke polynomial.* For $p \le 31$, $\Psi_{N,p}$ is built modulo p^2 from power sums $T(q^p)^m + p\,(T^m|U_p)$ and Newton's identities, computed modulo p^3. We verify the Kronecker congruence, the symmetry of $\overline{H}$, the identity $-\overline{H}(t^p,t) = \mathrm{Res}_t\, R_{N,p}$ at *all* $t \in \mathbb{F}_{p^2}$, and part (f).

(8) *Independent re-implementation* (`code/independent_check`, written from scratch by a separate AI audit pass over this note). For 10 pairs, including $(13,29)$, $(7,29)$ and $(5,37)$, it determines $\mathrm{den}_{\mathrm{red}}\, R_{N,p}$ by Berlekamp–Massey on the $1/T$-expansion of $\overline{T|U_p}$, and the supersingular values with their weights by factoring Δ_N^{-k} written in T. It uses no other supersingular input. It confirms $\mathrm{den}_{\mathrm{red}}\, R = D_{\mathrm{thin}}$ and $r_i W_i = -\left(\frac{-N}{p}\right)$. Berlekamp–Massey is run on a finite number of terms without a certificate, so this is a consistency check only. The same audit tested Lemma 6.1 numerically for $\ell = 11, 17, 19, 23$ and Conjecture 1.3 for $p \le 47$.

8.2 Results

Reproduction. For $N \in \mathcal{N}$ and $p \in \{5, \dots, 29\}$, $p \neq N$, there are 37 pairs with 135 thin points, of which 60 are non-rational. 26 pairs have at least two thin points, 7 have one and 4 have none. Every ratio is $-\left(\frac{-N}{p}\right)$, and at $p = 29$ the constants are $+1, +1, -1, -1, -1$ in level order. This matches the inherited report exactly. Table 2 gives the details.

Extension. Table 1 summarizes all runs. Tier 1 (certified) covers 222 pairs, all $p \le 211$, with 6,246 thin points (4,696 non-rational). Tier 2 covers 960 further pairs with $223 \le p \le 1499$ and 233,336 thin points (210,326 non-rational). In every pair the ratio set is a single value, equal to $-\left(\frac{-N}{p}\right)$. Every r_i and every W_i lies in $\mathbb{F}_p^\times$, and the Frobenius relations hold. The candidate-free denominator equals D_{thin} for all $p \le 37$. The Hecke-polynomial checks pass for all $p \le 31$. Theorem 1.1 holds at every supersingular point of $X_0(N)$, $N \in \mathcal{N}$, for all $p \le 1499$. Baker's congruence and the constancy of c hold for all $p \le 1499$ (within the per-pair runs) and, separately, for all $p \le 997$ at level one. Conjecture 1.3 holds for all 166 primes $5 \le p \le 997$. The exploratory level-11 run ($5 \le p \le 61$, $p \neq 11$) gave $-\left(\frac{-11}{p}\right)$ in all 15 cases.

Failures and exceptions. No counterexample to any stated identity was found. One check initially reported a failure: the Kaneko–Zagier polynomial identity for $(N,p) = (13,5)$. The cause was a bug in the check, which compared beyond the computed length of the q-series ($p^3 + 2 = 127$ terms). After the series was lengthened, the check passed; the residues themselves were unaffected.

N	range	tier	pairs	thin pts	non-rat. thin	$\#\{r_i W_i\} > 1$	all $= -\left(\frac{-N}{p}\right)$
2	$5 \le p \le 211$	Tier 1	45	530	310	0	yes
3	$5 \le p \le 211$	Tier 1	45	716	456	0	yes
5	$7 \le p \le 211$	Tier 1	44	1059	740	0	yes
7	$5 \le p \le 211$	Tier 1	44	1431	1116	0	yes
13	$5 \le p \le 211$	Tier 1	44	2510	2074	0	yes
2	$223 \le p \le 1499$	Tier 2	192	19978	16586	0	yes
3	$223 \le p \le 1499$	Tier 2	192	26669	22808	0	yes
5	$223 \le p \le 1499$	Tier 2	192	39960	35524	0	yes
7	$223 \le p \le 1499$	Tier 2	192	53338	48318	0	yes
13	$223 \le p \le 1499$	Tier 2	192	93391	87090	0	yes

Table 1: All runs by level. Tier 1 = certified reconstruction of $R_{N,p}$; Tier 2 = residues from the proven double-pole lemma. Every pair has a single ratio value, equal to $-\left(\frac{-N}{p}\right)$.

Table 2: Reproduction range $p \le 29$ (certified). $\#j_{ss}$: supersingular j; $\#x$: supersingular points of $X_0(N)$; s: supersingular T-values; weights written e^{count}; last two columns: observed ratio set and the predicted constant.

N	p	$\#j_{ss}$	$\#x$	s	weights e	thin	non-rat.	$\{r_i W_i\}$	$-\left(\frac{-N}{p}\right)$
2	5	1	1	1	2^1	0	0	$--$	$+1$
3	5	1	2	2	$2^1, 6^1$	0	0	$--$	$+1$
7	5	1	4	3	$2^2, 3^1$	0	0	$--$	$+1$
13	5	1	6	4	$1^1, 2^2, 3^1$	1	0	$\{+1\}$	$+1$
2	7	1	2	2	$2^1, 4^1$	0	0	$--$	$+1$
3	7	1	2	1	1^1	1	0	$\{-1\}$	-1
5	7	1	4	2	$1^1, 2^1$	1	0	$\{-1\}$	-1
13	7	1	8	4	$1^3, 2^1$	3	2	$\{-1\}$	-1
2	11	2	3	2	$1^1, 4^1$	1	0	$\{-1\}$	-1
3	11	2	4	3	$1^1, 2^1, 6^1$	1	0	$\{+1\}$	$+1$
5	11	2	6	4	$1^1, 2^3$	1	0	$\{+1\}$	$+1$
7	11	2	8	4	$1^3, 3^1$	3	2	$\{-1\}$	-1
13	11	2	14	7	$1^5, 2^1, 3^1$	5	2	$\{-1\}$	-1
2	13	1	3	2	$1^1, 2^1$	1	0	$\{+1\}$	$+1$
3	13	1	4	2	1^2	2	0	$\{-1\}$	-1
5	13	1	6	4	$1^2, 2^2$	2	0	$\{+1\}$	$+1$
7	13	1	8	5	$1^3, 2^2$	3	2	$\{+1\}$	$+1$
2	17	2	4	2	1^2	2	0	$\{-1\}$	-1
3	17	2	6	4	$1^2, 2^1, 6^1$	2	0	$\{+1\}$	$+1$
5	17	2	8	5	$1^3, 2^2$	3	2	$\{+1\}$	$+1$
7	17	2	12	7	$1^4, 2^2, 3^1$	4	0	$\{+1\}$	$+1$
13	17	2	20	10	$1^9, 3^1$	9	4	$\{-1\}$	-1
2	19	2	5	3	$1^2, 4^1$	2	0	$\{-1\}$	-1
3	19	2	6	3	1^3	3	2	$\{-1\}$	-1
5	19	2	10	6	$1^3, 2^3$	3	0	$\{+1\}$	$+1$
7	19	2	12	7	$1^5, 2^2$	5	4	$\{+1\}$	$+1$
13	19	2	22	11	$1^{10}, 2^1$	10	8	$\{-1\}$	-1
2	23	3	6	4	$1^2, 2^1, 4^1$	2	0	$\{+1\}$	$+1$
3	23	3	8	5	$1^3, 2^1, 6^1$	3	0	$\{+1\}$	$+1$
5	23	3	12	6	$1^5, 2^1$	5	4	$\{-1\}$	-1
7	23	3	16	8	$1^7, 3^1$	7	4	$\{-1\}$	-1
13	23	3	28	15	$1^{11}, 2^3, 3^1$	11	4	$\{+1\}$	$+1$
2	29	3	7	4	$1^3, 2^1$	3	2	$\{+1\}$	$+1$
3	29	3	10	6	$1^4, 2^1, 6^1$	4	0	$\{+1\}$	$+1$
5	29	3	14	7	1^7	7	2	$\{-1\}$	-1
7	29	3	20	10	$1^9, 3^1$	9	6	$\{-1\}$	-1
13	29	3	34	17	$1^{16}, 3^1$	16	10	$\{-1\}$	-1

Small characteristic. Table 3 lists all non-thin points for $p \le 13$. Two features are worth noting. First, the w_N-fixed points are not always the reductions of the characteristic-zero elliptic points one might expect. For $(N, p) = (2, 7)$, the CM point $j = 8000$ reduces to $j = 1728$, and the Fricke-fixed point $u = 64$ merges with the double point $u = 512$ over 1728. The resulting point has $(m_x, |\operatorname{Stab}|) = (1, 2)$ and $e = 2$, while $u = -64$ has $(2, 2)$ and $e = 4$. Second, for $(N, p) = (3, 5)$ a point with $e = 6$ occurs, for which $(p+1)/e = 1$. In all cases the characteristic-p definition of e is what the residue formula and Proposition 4.3 require.

Table 3: All non-thin supersingular values for $p \leq 13$. Elements of $\mathbb{F}_{p^2} \setminus \mathbb{F}_p$ are written in python-flint's basis z (minimal polynomial recorded per prime in the JSON output).

N	p	t	u over t	j	$(m_x, \lvert\mathrm{Stab}\rvert)$	e
2	5	2	4	0	$(1,2)$	2
3	5	3	3	0	$(3,2)$	6
3	5	1	2	0	$(1,2)$	2
7	5	1	$z+3, 4z+4$	0	$(3,1)$	3
7	5	0	3	0	$(1,2)$	2
7	5	3	2	0	$(1,2)$	2
13	5	2	$4z+3, z+2$	0	$(3,1)$	3
13	5	$z+4$	$3z+1$	0	$(1,2)$	2
13	5	$4z$	$2z+4$	0	$(1,2)$	2
2	7	1	6	6	$(2,2)$	4
2	7	5	1	6	$(1,2)$	2
5	7	5	$5z+4, 2z+2$	6	$(2,1)$	2
13	7	3	$2z+3, 5z+5$	6	$(2,1)$	2
2	11	6	2	1	$(2,2)$	4
3	11	0	5	1	$(1,2)$	2
3	11	2	6	0	$(3,2)$	6
5	11	6	$3z+5, 8z+6$	1	$(2,1)$	2
5	11	10	2	1	$(1,2)$	2
5	11	2	9	0	$(1,2)$	2
7	11	2	$3z+4, 8z+5$	0	$(3,1)$	3
13	11	7	$3z+2, 8z+3$	1	$(2,1)$	2
13	11	8	$3z+8, 8z+9$	0	$(3,1)$	3
2	13	9	12	5	$(1,2)$	2
5	13	$2z+5$	$z+6$	5	$(1,2)$	2
5	13	$11z+7$	$12z+7$	5	$(1,2)$	2
7	13	5	7	5	$(1,2)$	2
7	13	3	6	5	$(1,2)$	2

9 Audit of the inherited level-two material

- *Normalization.* The level-two manuscript (Jordan, 2026) uses the same $T_{2A} = u + 24 + 4096/u$, the same U_p and residue convention, and weights $e = 4$ at $T = -104$ ($u = -64$: $m_x = 2$, fixed) and $e = 2$ at $T = 152$ ($u = 64$: $m_x = 1$, fixed). These agree with the general definition. Its Theorem 2.2 is our Proposition 3.2 for $N = 2$.

- *Its proof.* The manuscript and the handoff note prove the $N = 2$ case by descent from the Legendre modular polynomial. The steps are a Kaneko–Zagier-type correction at Legendre level, a crystalline lattice lemma giving $W(x)^{p-1} = -s(x)$, the twist argument $s(x) = \left(\frac{-1}{p}\right)$ (Auer and Top, 2002, Prop. 2.2), and the norm identity built on Auer and Top (2002, Prop. 3.1). We did not use any of these. Our proof for $N = 2$ is independent and reaches the same constant $-\left(\frac{-2}{p}\right)$. The crystalline inputs of that note (its Fact I3) are therefore not needed for the present results.

- *Claims checked.* The handoff states that Baker settled the j-level sign. We confirmed Baker's Theorem 1.4 from the author's copy. Its statement that Betina and Lecouturier (2017, Thm. 1.1(iii)) rests on a torsion cancellation that fails for $p \equiv 3 \pmod 4$ concerns the Legendre-level period theorem, which we do not use.

- *Level-two manuscript.* Unchanged, as requested.

10 Literature assessment

Sources read in this investigation are marked (R); sources known only from abstracts or from statements quoted elsewhere are marked (A).

- de Shalit (1994) (R): level one. For supersingular $j_i \notin \{0, 1728\}$, Theorem 1 and Corollary 3 give
$$\overline{R}(j_i) = \pm(-1)^{(p-1)/2} \prod_{k \neq i} (j_i - j_k)^{-(p+1)/\delta_k},$$
with $+$ if $p \equiv 1 \pmod 4$ or $j_i \in \mathbb{F}_p$, where $R(X) = F_p(X, X^p)/p$; also $\overline{R}(j) = 0$ at ordinary j and at $j \in \{0, 1728\}$. The proof is via p-adic periods and the minimal regular model. In our normalization this is $r_i W_i = -\left(\frac{-1}{p}\right)$ up to sign, i.e. "$N = 1$".

- Kaneko and Zagier (1998, §10) (R): level one. Elementary derivation of the analogue of Lemma 5.1, of the pole support, and of $H_p(j) \equiv -(\cdots)/S_p'(j)^2$. They show de Shalit's signed formula (39) is equivalent to (40) and ask for an elementary proof.

- Baker (1998) (R, author's copy): proves (40), hence the signed level-one formula, for all $p \geq 5$.

- Betina and Lecouturier (2017) (R) and Betina and Lecouturier (2018) (R, in the handoff): Legendre level. They state the analogue of de Shalit's formula up to a sign for $p \equiv 3 \pmod 4$, $\lambda \notin \mathbb{F}_p$ (BL18, Cor. 1.2). The handoff's argument fixes the sign at $+$. Neither treats $\Gamma_0(N)^+$.

- Jordan (2026) and its handoff (R): level two only, AI-assisted, not refereed.

- Ono (2025) (R via arXiv): $\Delta^{(p^2-1)/12}$ and $(\Delta(\ell\tau)/\Delta(\tau))^{(p^2-1)/12}$ at supersingular points. Theorem 1.1 refines the latter.

- Morton (2021), Nakaya (2019), Sakai (2011) (A): supersingular polynomials for Fricke groups (linear factors, resultant identities, Atkin polynomials). None of these is about U_p residues.

- Chen et al. (2019) (A): p-adic annihilation of Hauptmoduln; no residue formulas. Bringmann et al. (2026) (A): level one, meromorphic forms; signs not explicit.

We found no published statement of the Main Theorem for $N > 1$. It does not appear in the sources above, and targeted web searches turned up nothing further. This is not evidence of novelty: the searches were limited, and the final journal text of Betina and Lecouturier (2019) was not inspected. Corollary 1.2 and Conjecture 1.3 were also not found, but the literature around Baker's paper was not searched systematically, in particular Robert's work on congruences of Eisenstein series and the homotopy-theoretic work of Behrens and Larson on the divided beta family. Their novelty should be checked there.

11 What is proved, and what remains to be checked

Proved here, given Facts I1–I7. Main Theorem (a)–(f) for $N \in \{2, 3, 5, 7, 13\}$ and all primes $p \geq 5$, $p \neq N$; Theorem 1.1 for all primes $\ell \neq p$; Corollary 1.2.

Steps a specialist should check.

(U1) Lemma 4.1(ii) and the order formula in Lemma 5.2. These use Kodaira–Spencer on a fine moduli cover and the simple zeros of the Hasse invariant (Facts I2(ii),(iv)). Both are standard. The resulting divisor statements are confirmed exactly by Proposition 4.3 and by the orders of $(\theta T)^{p-1} D^2_{\text{thin}}$ for $p \le 211$, but this is not a proof.

(U2) The reduction mod p of the Serre-derivative identity in Lemma 5.2 ($\partial A = B/12$ at supersingular points), and Lemma 5.3. These are short q-expansion-principle arguments; the numerical identity $r_i W_i = \left(\frac{N}{p}\right) c(E)c(E')$ confirms the combination.

(U3) Theorem 1.1. This relies on the moduli interpretation of diamond operators and Frobenius on $X_1(576\ell)$ (Fact I5) and on the Gordon–Hughes–Newman theorem (Fact I4). The multiplier is verified exactly. Two convention-sensitive points are harmless: the direction of $\langle \pm p^{\pm 1} \rangle$, because the character is even and quadratic; and the field of definition of G, which is $\mathbb{Q}$ in the μ_M-embedding model and $\mathbb{Q}(\sqrt{\ell})$ in the point model, where only the p^2-Frobenius is used.

(U4) Fact I3 is imported from Baker (1998); we did not re-derive it.

(U5) Fact I1 is proved only in outline (Proposition 3.1), following an unpublished manuscript.

(U6) Fact I2(v): the special fibre of $X_0(N)^+$ is $\mathbb{P}^1$ with coordinate $\overline{T}$. For the five levels this is elementary; for other genus-zero Fricke levels it is imported.

Not claimed. Conjecture 1.3; any statement about the final TAMS text of Betina and Lecouturier (2019); the level-11 computations as anything more than exploratory.

AI-assistance declaration. This note, its proofs and its code were produced by an AI system (Claude, Anthropic) at the request of D. Jordan. A separate AI audit pass reviewed the proofs. It found no error that breaks the argument, and its corrections (precision of the moduli conventions in Theorem 1.1, scope of Lemmas 4.1 and 5.3, an inverted ratio and several wording issues) are incorporated. This is not a substitute for review by a specialist, which has not taken place.

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